Programming Fundamentals Using C++

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Week 03

Lecture – 01

Lecture – 02

Building Programs: Basic input/output

Semester Calendar

Class Schedule

Lab Hours (Practice)

Tuesday (09:00 to 10:30)
Wednesday (09:30 to 12:30)
Wednesdy (03:00 to 04:30)

Basic Input/Output

- The example programs of the previous sections printed simple values on screen, but the standard library provides many additional ways to interact with the user via its input/output features.
- C++ uses a convenient abstraction called streams to perform input and output operations
- C++ comes with libraries that provide us with many ways for performing input and output.

Basic Input/Output:

- **Input Stream:** If the direction of flow of bytes is from the device (for example, Keyboard) to the main memory then this process is called input.
- **Output Stream:** If the direction of flow of bytes is opposite, i.e. from main memory to device(display screen) then this process is called output.
- **cin, cout, cerr,** and **clog** are streams that handle standard inputs and standard outputs. These are stream objects defined in iostream header file.

stream	description
cin	standard input stream
cout	standard output stream
cerr	standard error (output) stream
clog	standard logging (output) stream

Standard output (cout)

- On most program environments, the standard output by default is the screen, and the C++ stream object defined to access it is cout

```
cout << "Output sentence";          // prints Output sentence on screen
cout << 120;                        // prints number 120 on screen
cout << x;                          // prints the value of x on screen
```

- Multiple insertion operations (<<) may be chained in a single statement:

```
cout << "This " << " is a " << "single C++ statement";
```

- Chaining insertions is especially useful to mix literals and variables in a single statement:

```
cout << "I am " << age << " years old and my zipcode is " << zipcode;
```

Standard input (cin):

```
int age;
cin >> age;
```

- The first statement declares a variable of type int called age,
- the second statement extracts from cin a value to be stored in it.
- This operation makes the program wait for input from cin; generally, this means that the program will wait for the user to enter some sequence with the keyboard

Program to accept values at Run Time (Using cin>>)

```
//cin>> Example
#include <iostream>
using namespace std;
int main ()
{
  int i;
  cout << "Please enter an integer value: ";
  cin >> i;
  cout << "The value you entered is " << i;
  return 0;        }
```

Cin>>

```
cin >> a;
cin >> b;
```

- Extractions on cin can also be chained to request more than one datum in a single statement:

```
cin >> a >> b;
```

Using Static variables

```
// using Static Variables
#include <iostream>
using namespace std;
int main()
{
  int a, b, c;
  a = 3; b = 4;
  int c = a + b ;
  cout << "Ans is “<<c;
   return 0;  }
```

Values at runtime (using cin>>)

```
// using cin>>
#include <iostream>
using namespace std;
int main()
{
    int a, b;
    cin>>a;   cin>>b;
    int c = a + b  ;
    cout << "Ans is “<<c;
    return 0; }
```

using cerr & clog

Program 1

```
// using Cerr
#include <iostream>
using namespace std;
int main()
{
    cerr << "An error occured";
    return 0;
}
```

Program 2

```
// using Clog
#include <iostream>
using namespace std;
int main()
{
    clog << "An error occured";
    return 0;
}
```

Any Question